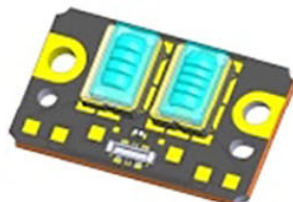


Part No. **NUMB12**

■Features

- High Power Multiple Laser Diode (LD) Chip Package
- A common cathode for 9 chips to emit collimated beams
- Blue 2 series, Green 3 series, Red 4 series
- High Heat Dissipation
- Pulse operation



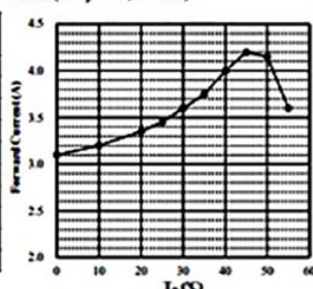
■Absolute Maximum Ratings

Item	Symbol	Color	Absolute Maximum Ratings	Unit
Forward Current	I_f	Blue	3.5	A
		Green	2.3	A
		Red	Table 1	A
Cathode Current (CW) ¹	I_c	Multiple	5.0	A
Allowable Reverse Current	I_R	Blue	85	mA
		Green	85	mA
		Red	85	mA
Storage Temperature	T_{stg}	-	-40 ~ 85	°C
Operating Temperature	T_m	-	0 ~ 55	°C

¹ When two or more different colored chips are operated simultaneously.

Table 1 Derating characteristics of Red (Duty 40%, 240Hz)

T_m (°C)	Forward Current (A)
0	3.10
10	3.20
20	3.35
25	3.45
30	3.60
35	3.75
40	4.00
45	4.20
50	4.15
55	3.60



■Initial Electrical/Optical Characteristics

Item	Symbol	Color	Condition ²	Min	Typ.	Max	Unit
Optical Output Power	P_o	Blue	CW	7.1	(8.3)	9.5	W
		Green		4.0	(4.8)	5.5	W
		Red		8.5	(10.0)	11.5	W
Dominant Wavelength	λ_d	Blue	CW	458	(465)	472	nm
		Green		519	(525)	531	nm
		Red ³		635	(639) (643) (647)	651	nm
Threshold Current	I_{th}	Blue	CW	300	-	550	mA
		Green		150	-	500	mA
		Red		400	-	650	mA
Slope Efficiency	η	Blue	CW	-	(3.2)	-	W/A
		Green		-	(2.5)	-	W/A
		Red		-	(4.0)	-	W/A
Forward Voltage	V_f	Blue	CW	7.0	(8.4)	10.0	V
		Green		12.5	(14.5)	17.5	V
		Red		7.5	(9.0)	11.0	V
Beam Pointing Tilt Angle ^{4,5}	$\Delta\theta$	Blue	CW	-	-	1.2	°
		Green		-	-	1.2	°
		Red		-	-	1.2	°
Beam Divergence ⁵	Parallel	Blue	CW	0	(0.8)	1.6	°
		Green		-0.4	(0.4)	1.2	°
		Red		4.0	(5.0)	6.0	°
	Perpendicular	Blue	CW	-1.5	(0)	1.5	°
		Green		-1.5	(0)	1.5	°
		Red		-1.5	(0)	1.5	°

() are reference figures.

² The characteristics were measured per color using the following forward current:

Blue: $I_f=3.0A$, Green: $I_f=2.1A$, Red: $I_f=3.0A$

³ Composed of 3 ranges.

⁴ $\Delta\theta = \sqrt{\Delta\theta_x^2 + \Delta\theta_y^2}$ ($\Delta\theta$ is the value of the total emission per color.)

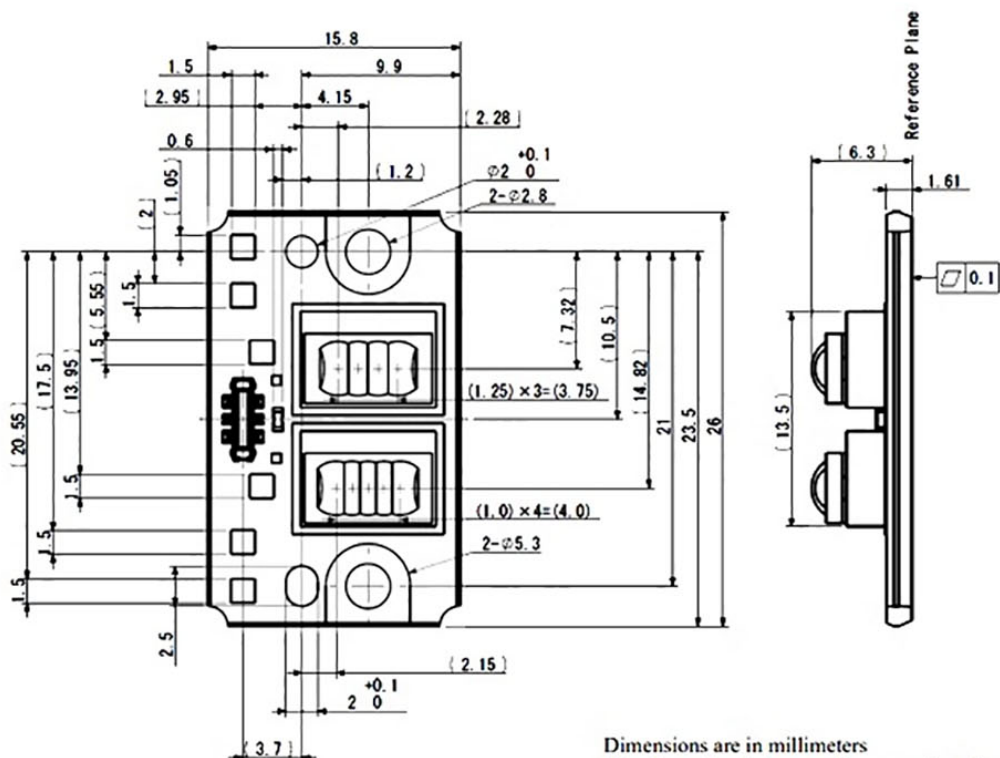
⁵ Full angle at 1/e² of peak intensity for the total emission per color.

All figures in this specification are measured by Nichia's method and may contain measurement deviations.

The above specifications are for reference purpose only and subjected to change without prior notice.

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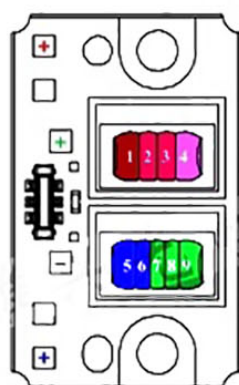
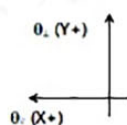
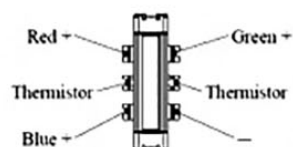
Outline Dimensions



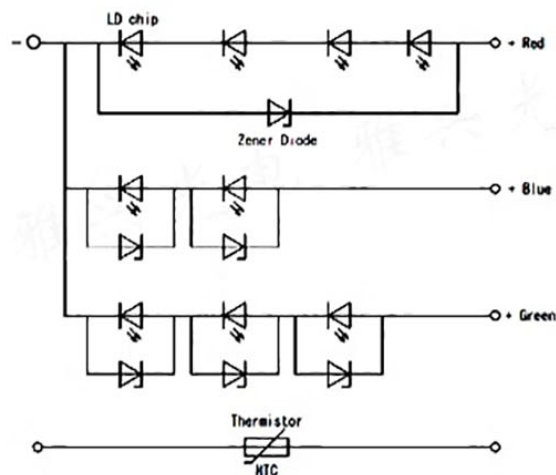
※Note
TOLERANCES based on JIS B 0405-m(Table), UNLESS NOTED
Table, Limits for Linear measures(mm)

Range	0.5 to 3	above 3 to 6	above 6 to 30
Tolerance	±0.1	±0.1	±0.2

LD chips Position No. & Drive Circuit



- Red(647nm) Pos. 1
- Red(643nm) Pos. 2, 3
- Red(639nm) Pos. 4
- Blue(465nm) Pos. 5, 6
- Green(525nm) Pos. 7, 8, 9



This model has a Zener Diode built in as a protection circuit against static electricity.